

CODING :

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>SmartExam — National Exam Slot Booking & Result Management</title>

<style>

*{box-sizing:border-box;margin:0;padding:0}

:root{

  --primary:#2563eb;--primary2:#1d4ed8;--dark:#0f172a;--text:#334155;

  --muted:#64748b;--bg:#f5f7fb;--card:#fff;--border:#e2e8f0;

  --success:#16a34a;--warning:#d97706;--danger:#dc2626;--purple:#7c3aed;

}

body{font-family:Inter,Segoe UI,Arial,sans-serif;background:var(--bg);color:var(--text);line-height:1.55}

button,input,select{font:inherit}

button{cursor:pointer}

a{text-decoration:none;color:inherit}

.app{display:flex;min-height:100vh}

.sidebar{width:260px;background:#0b1220;color:#cbd5e1;position:fixed;inset:0 auto 0 0;z-index:20;display:flex;flex-direction:column}

.brand{padding:25px 22px;border-bottom:1px solid #1e293b;display:flex;align-items:center;gap:12px}

.logo{width:40px;height:40px;border-radius:12px;background:linear-gradient(135deg,#3b82f6,#7c3aed);display:grid;place-items:center;color:white;font-weight:900}

.brand h2{font-size:19px;color:#fff}.brand small{display:block;color:#94a3b8;font-size:10px}

.nav{padding:18px 12px;overflow:auto}.nav-section{font-size:10px;text-transform:uppercase;color:#64748b;font-weight:800;padding:12px 12px 7px;letter-spacing:1px}

.nav button{width:100%;background:none;border:0;color:#cbd5e1;text-align:left;padding:11px 13px;border-radius:10px;margin:2px 0;font-size:14px}
```

```

.nav button:hover,.nav button.active{background:#1e3a8a;color:#fff}

.nav button span{width:24px;display:inline-block}

.profile{margin-top:auto;padding:15px;border-top:1px solid #1e293b}

.profile-box{background:#111c30;padding:12px;border-
radius:12px}.avatar{width:36px;height:36px;background:#dbeafe;color:#1d4ed8;border-
radius:50%;display:grid;place-items:center;font-weight:800}

.profile-row{display:flex;align-items:center;gap:10px}.profile strong{font-
size:13px;color:white}.profile small{font-size:11px;color:#94a3b8}

.main{margin-left:260px;width:calc(100% - 260px)}

.topbar{height:72px;background:white;border-bottom:1px solid var(--border);display:flex;align-
items:center;justify-content:space-between;padding:0 32px;position:sticky;top:0;z-index:10}

.breadcrumb{font-size:13px;color:var(--muted)}.breadcrumb b{color:var(--dark)}

.top-actions{display:flex;align-items:center;gap:10px}

.icon-btn{width:38px;height:38px;border:1px solid var(--border);background:#fff;border-radius:10px}

.content{padding:30px;max-width:1500px;margin:auto}

.page{display:none}.page.active{display:block}

.page-head{display:flex;justify-content:space-between;align-items:center;margin-bottom:24px}.page-
head h1{font-size:26px;color:var(--dark)}.page-head p{color:var(--muted);font-size:14px;margin-
top:3px}

.btn{border:0;border-radius:9px;padding:10px 16px;font-weight:700;font-size:13px;transition:.2s}

.btn-primary{background:var(--primary);color:#fff}.btn-primary:hover{background:var(--primary2)}

.btn-light{background:#f1f5f9;color:#334155}.btn-danger{background:#fee2e2;color:#b91c1c}.btn-
success{background:#dcfce7;color:#15803d}.btn-outline{background:#fff;color:var(--
primary);border:1px solid #bfdbfe}

.hero{background:linear-gradient(120deg,#172554,#2563eb 60%,#4f46e5);border-
radius:20px;padding:30px;color:white;display:flex;justify-content:space-between;align-
items:center;overflow:hidden;position:relative;margin-bottom:25px}

.hero:after{content:"";position:absolute;width:280px;height:280px;border-
radius:50%;background:#ffffff12;right:-70px;top:-100px}

.hero h1{font-size:30px;max-width:700px}.hero p{margin:10px 0 20px;color:#dbeafe;max-
width:700px;font-size:14px}

.hero-actions{display:flex;gap:10px}.hero .btn-light{background:#fff;color:#1d4ed8}.hero .btn-
outline{border-color:#93c5fd;background:transparent;color:white}

```

```
.grid{display:grid;gap:18px}.grid-4{grid-template-columns:repeat(4,1fr)}.grid-3{grid-template-columns:repeat(3,1fr)}.grid-2{grid-template-columns:repeat(2,1fr)}
```

```
.stat{background:#fff;border:1px solid var(--border);border-radius:15px;padding:19px;display:flex;justify-content:space-between;align-items:center;box-shadow:0 2px 8px #0f172a06}
```

```
.stat .label{font-size:12px;color:var(--muted);font-weight:700}.stat .value{font-size:27px;color:var(--dark);font-weight:800;margin-top:3px}.stat-icon{width:45px;height:45px;border-radius:12px;display:grid;place-items:center;font-size:20px}
```

```
.blue{background:#dbeafe;color:#2563eb}.green{background:#dcfce7;color:#16a34a}.orange{background:#ffedd5;color:#ea580c}.purple{background:#ede9fe;color:#7c3aed}
```

```
.card{background:#fff;border:1px solid var(--border);border-radius:15px;box-shadow:0 2px 8px #0f172a06}.card-head{padding:17px 19px;border-bottom:1px solid var(--border);display:flex;justify-content:space-between;align-items:center}.card-head h3{font-size:15px;color:var(--dark)}.card-body{padding:19px}
```

```
.table-wrap{overflow:auto}table{width:100%;border-collapse:collapse;font-size:13px}th{background:#f8f9fa;color:#6c757d;font-size:11px;text-transform:uppercase;letter-spacing:.4px;text-align:left;padding:12px;border-bottom:1px solid var(--border)}td{padding:13px 12px;border-bottom:1px solid #e2e3e5;color:#495052}tr:last-child td{border-bottom:0}
```

```
.badge{display:inline-flex;padding:4px 8px;border-radius:20px;font-size:10px;font-weight:800}.badge-green{background:#dcfce7;color:#28a745}.badge-orange{background:#ffedd5;color:#fd7e14}.badge-red{background:#fee2e2;color:#dc3545}.badge-blue{background:#dbeafe;color:#17a2b8}.badge-purple{background:#ede9fe;color:#6f42c1}
```

```
.progress{height:7px;background:#e2e3e5;border-radius:20px;overflow:hidden}.progress i{display:block;height:100%;background:#2563eb;border-radius:20px}
```

```
.exam-card{padding:20px}.exam-icon{width:48px;height:48px;background:#dbeafe;color:#2563eb;border-radius:13px;display:grid;place-items:center;font-size:21px;margin-bottom:13px}.exam-card h3{color:var(--dark);font-size:17px}.exam-card p{font-size:12px;color:var(--muted);margin:6px 0 15px}.meta{display:flex;gap:14px;flex-wrap:wrap;font-size:11px;color:#64748b;margin-bottom:16px}.meta b{color:#334155}
```

```
.slot{padding:18px}.slot-top{display:flex;justify-content:space-between;gap:10px}.slot h3{font-size:16px;color:var(--dark)}.slot .center{font-size:12px;color:var(--muted);margin:5px 0 13px}.slot-info{background:#f8f9fa;border-radius:10px;padding:11px;display:grid;grid-template-columns:repeat(3,1fr);gap:8px;margin-bottom:14px}.slot-info div{font-size:11px;color:#64748b}.slot-info b{display:block;color:#334155;font-size:12px;margin-top:2px}
```

```
.empty{text-align:center;padding:40px;color:#94a3b8}.search{border:1px solid var(--border);border-radius:9px;padding:9px 12px;outline:0;width:230px}.search:focus,input:focus,select:focus{border-color:#93c5fd;box-shadow:0 0 3px #dbeafe}
```

```

.form-grid{display:grid;grid-template-columns:repeat(2,1fr);gap:15px}.field label{display:block;font-size:12px;font-weight:700;color:#475569;margin-bottom:6px}.field input,.field select,.field textarea{width:100%;border:1px solid var(--border);border-radius:9px;padding:10px 11px;background:#fff;outline:0}.field textarea{min-height:90px;resize:vertical}.full{grid-column:1/-1}

.modal{position:fixed;inset:0;background:#0f172a88;display:none;place-items:center;z-index:100;padding:20px}.modal.show{display:grid}.modal-box{background:#fff;border-radius:17px;width:min(620px,100%);max-height:90vh;overflow:auto}.modal-head{padding:18px 20px;border-bottom:1px solid var(--border);display:flex;justify-content:space-between}.modal-body{padding:20px}.close{background:none;border:0;font-size:22px;color:#64748b}

.toast{position:fixed;right:25px;bottom:25px;background:#0f172a;color:#fff;padding:13px 17px;border-radius:10px;box-shadow:0 10px 30px #0003;display:none;z-index:200;font-size:13px}.toast.show{display:block}

.question{padding:19px;border:1px solid var(--border);border-radius:13px;background:#fff;margin-bottom:15px}.question h3{font-size:14px;color:var(--dark);margin-bottom:12px}.option{display:block;padding:10px;border:1px solid var(--border);border-radius:8px;margin:7px 0;cursor:pointer;font-size:13px}.option:hover{background:#f8fafc}.option input{margin-right:8px}

.kpi{font-size:12px;color:var(--muted)}.kpi strong{display:block;font-size:23px;color:var(--dark)}

.code{background:#0b1220;color:#dbeafe;padding:17px;border-radius:12px;font:12px Consolas,monospace;white-space:pre-wrap;overflow:auto}

.hash-bucket{border:1px dashed #cbd5e1;border-radius:12px;padding:13px;min-height:100px}.hash-bucket h4{font-size:12px;margin-bottom:9px}.chain{padding:8px;background:#eff6ff;border-radius:8px;font-size:11px;margin:5px 0}

.mobile-menu{display:none}

@media(max-width:1000px){.grid-4{grid-template-columns:repeat(2,1fr)}.grid-3{grid-template-columns:1fr 1fr}}

@media(max-width:760px){.sidebar{transform:translateX(-100%);transition:.25s}.sidebar.open{transform:translateX(0)}.main{margin-left:0;width:100%}.mobile-menu{display:block}.content{padding:18px}.topbar{padding:0 18px}.grid-4,.grid-3,.grid-2,.form-grid{grid-template-columns:1fr}.hero{padding:22px}.hero h1{font-size:23px}.hero-visual{display:none}.search{width:150px}}
</style>
</head>
<body>
<div class="app">
<aside class="sidebar" id="sidebar">

```

```
<div class="brand"><div class="logo">SE</div><div><h2>SmartExam</h2><small>NATIONAL EXAM  
PLATFORM</small></div></div>
```

```
<nav class="nav">
```

```
<div class="nav-section">Main</div>
```

```
<button class="active" onclick="showPage('dashboard',this)"><span>■</span>  
Dashboard</button>
```

```
<button onclick="showPage('exams',this)"><span>■</span> Examinations</button>
```

```
<button onclick="showPage('slots',this)"><span>☐</span> Slot Booking</button>
```

```
<button onclick="showPage('bookings',this)"><span>✓</span> My Bookings</button>
```

```
<button onclick="showPage('attempt',this)"><span>🔍</span> Exam Attempt</button>
```

```
<button onclick="showPage('results',this)"><span>Ⓢ</span> Results</button>
```

```
<div class="nav-section">DBMS</div>
```

```
<button onclick="showPage('analytics',this)"><span>📊</span> DBMS Analytics</button>
```

```
<button onclick="showPage('hashing',this)"><span>#</span> Hashing Demo</button>
```

```
<button onclick="showPage('transactions',this)"><span>🔄</span> Transactions</button>
```

```
<div class="nav-section">Administration</div>
```

```
<button onclick="showPage('centers',this)"><span>📍</span> Exam Centers</button>
```

```
<button onclick="showPage('users',this)"><span>👤</span> Users & Roles</button>
```

```
</nav>
```

```
<div class="profile"><div class="profile-box"><div class="profile-row"><div  
class="avatar">HK</div><div><strong>Harshith  
Karnam</strong><small>Administrator</small></div></div></div></div>
```

```
</aside>
```

```
<div class="main">
```

```
<header class="topbar"><div class="breadcrumb"><button class="icon-btn mobile-menu"  
onclick="toggleSide()">☰</button> <b>Smart National Exam System</b> <span> /  
Dashboard</span></div><div class="top-actions"><button class="icon-btn" onclick="notify('No new  
notifications')">🔔</button><button class="icon-btn" onclick="notify('System is  
online')">🟢</button></div></header>
```

```
<main class="content">
```

```
<section id="dashboard" class="page active">
```

```
<div class="hero">
```

```
<div><h1>Smart National Exam Slot Booking & Result Management</h1><p>Centralized
examination management with secure slot booking, indexed retrieval, transaction processing,
concurrency control and result management.</p><div class="hero-actions"><button class="btn btn-
light" onclick="showPage('slots')">Book an Exam Slot →</button><button class="btn btn-outline"
onclick="showPage('analytics')">View DBMS Analytics</button></div></div>
```

```
</div>
```

```
<div class="grid grid-4" style="margin-bottom:20px">
```

```
<div class="stat"><div><div class="label">Registered Candidates</div><div
class="value">12,480</div></div><div class="stat-icon blue">👤</div></div>
```

```
<div class="stat"><div><div class="label">Active Examinations</div><div
class="value">24</div></div><div class="stat-icon purple">📅</div></div>
```

```
<div class="stat"><div><div class="label">Available Slots</div><div
class="value">186</div></div><div class="stat-icon green">🪑</div></div>
```

```
<div class="stat"><div><div class="label">Published Results</div><div
class="value">9,742</div></div><div class="stat-icon orange">📄</div></div>
```

```
</div>
```

```
<div class="grid grid-2">
```

```
<div class="card"><div class="card-head"><h3>Recent Slot Bookings</h3><button class="btn btn-
light" onclick="showPage('bookings')">View All</button></div><div class="table-
wrap"><table><thead><tr><th>Candidate</th><th>Examination</th><th>Date</th><th>Status</th><
/tr></thead><tbody>
```

```
<tr><td>Arun Kumar</td><td>National Aptitude Exam</td><td>10 Sep 2026</td><td><span
class="badge badge-green">CONFIRMED</span></td></tr>
```

```
<tr><td>Priya S</td><td>CS Fundamentals</td><td>11 Sep 2026</td><td><span class="badge
badge-green">CONFIRMED</span></td></tr>
```

```
<tr><td>Rahul M</td><td>National Aptitude Exam</td><td>12 Sep 2026</td><td><span
class="badge badge-orange">PENDING</span></td></tr>
```

```
<tr><td>Divya R</td><td>CS Fundamentals</td><td>13 Sep 2026</td><td><span class="badge
badge-green">CONFIRMED</span></td></tr>
```

```
</tbody></table></div></div>
```

```

<div class="card"><div class="card-head"><h3>System Performance</h3><span class="badge badge-blue">LIVE</span></div><div class="card-body">
  <div style="margin-bottom:17px"><div style="display:flex;justify-content:space-between;font-size:12px;margin-bottom:6px"><b>Database Query Efficiency</b><span>94%</span></div><div class="progress"><i style="width:94%"></i></div></div>
  <div style="margin-bottom:17px"><div style="display:flex;justify-content:space-between;font-size:12px;margin-bottom:6px"><b>Slot Availability</b><span>78%</span></div><div class="progress"><i style="width:78%"></i></div></div>
  <div><div style="display:flex;justify-content:space-between;font-size:12px;margin-bottom:6px"><b>Transaction Success Rate</b><span>99.8%</span></div><div class="progress"><i style="width:99.8%"></i></div></div>
</div></div>
</div>
</section>

```

```

<section id="exams" class="page">

```

```

  <div class="page-head"><div><h1>Examinations</h1><p>Manage national-level examination programs and assessments.</p></div><button class="btn btn-primary" onclick="openModal('examModal')">+ Create Examination</button></div>

```

```

  <div class="grid grid-3">

```

```

    <div class="card exam-card"><div class="exam-icon">⌘</div><h3>National Aptitude Examination</h3><p>Comprehensive aptitude assessment covering logical reasoning, quantitative ability and verbal skills.</p><div class="meta"><span>⌚ <b>60 min</b></span><span>⌚ <b>10 questions</b></span><span>👤 <b>8,240 candidates</b></span></div><button class="btn btn-primary" onclick="startExam(1)">Start / Preview</button></div>

```

```

    <div class="card exam-card"><div class="exam-icon" style="background:#ede9fe;color:#7c3aed">⌘</div><h3>Computer Science Fundamentals</h3><p>Assessment covering database systems, operating systems, networking and computer fundamentals.</p><div class="meta"><span>⌚ <b>90 min</b></span><span>⌚ <b>30 questions</b></span><span>👤 <b>5,620 candidates</b></span></div><button class="btn btn-primary" onclick="notify('Examination preview opened')">Start / Preview</button></div>

```

```

    <div class="card exam-card"><div class="exam-icon" style="background:#dcfce7;color:#16a34a">⬢</div><h3>Digital Skills Assessment</h3><p>Evaluation of digital literacy, problem solving and technology awareness.</p><div class="meta"><span>⌚ <b>45 min</b></span><span>⌚ <b>25 questions</b></span><span>👤 <b>3,180 candidates</b></span></div><button class="btn btn-primary" onclick="notify('Examination preview opened')">Start / Preview</button></div>

```

</div>

</section>

<section id="slots" class="page">

<div class="page-head"><div><h1>Exam Slot Booking</h1><p>Select an examination center and reserve an available time slot.</p></div><input class="search" placeholder="Search slots..." oninput="filterSlots(this.value)"></div>

<div class="grid grid-2" id="slotGrid">

<div class="card slot" data-search="aptitude chennai central 10 september"><div class="slot-top"><div><h3>National Aptitude Examination</h3><div class="center">✦ Chennai Central Exam Centre · Chennai, Tamil Nadu</div></div>AVAILABLE</div><div class="slot-info"><div>Date10 Sep 2026</div><div>Time09:00 – 10:00</div><div>Seats76 / 100</div></div><button class="btn btn-primary" onclick="bookSlot('National Aptitude Examination','Chennai Central Exam Centre','10 Sep 2026','09:00 – 10:00')">Reserve Slot</button></div>

<div class="card slot" data-search="aptitude chennai central 10 september afternoon"><div class="slot-top"><div><h3>National Aptitude Examination</h3><div class="center">✦ Chennai Central Exam Centre · Chennai, Tamil Nadu</div></div>LIMITED</div><div class="slot-info"><div>Date10 Sep 2026</div><div>Time14:00 – 15:00</div><div>Seats8 / 100</div></div><button class="btn btn-primary" onclick="bookSlot('National Aptitude Examination','Chennai Central Exam Centre','10 Sep 2026','14:00 – 15:00')">Reserve Slot</button></div>

<div class="card slot" data-search="aptitude bengaluru 11 september"><div class="slot-top"><div><h3>National Aptitude Examination</h3><div class="center">✦ Bengaluru Knowledge Centre · Bengaluru, Karnataka</div></div>AVAILABLE</div><div class="slot-info"><div>Date11 Sep 2026</div><div>Time10:00 – 11:00</div><div>Seats54 / 80</div></div><button class="btn btn-primary" onclick="bookSlot('National Aptitude Examination','Bengaluru Knowledge Centre','11 Sep 2026','10:00 – 11:00')">Reserve Slot</button></div>

<div class="card slot" data-search="computer science hyderabad 13 september"><div class="slot-top"><div><h3>Computer Science Fundamentals</h3><div class="center">✦ Hyderabad National Test Centre · Hyderabad, Telangana</div></div>AVAILABLE</div><div class="slot-info"><div>Date13 Sep 2026</div><div>Time09:30 – 11:00</div><div>Seats63 / 90</div></div><button class="btn btn-primary" onclick="bookSlot('Computer Science Fundamentals','Hyderabad National Test Centre','13 Sep 2026','09:30 – 11:00')">Reserve Slot</button></div>

</div>

</section>

<section id="bookings" class="page">

<div class="page-head"><div><h1>My Bookings</h1><p>Track your reserved examination slots.</p></div><button class="btn btn-primary" onclick="showPage('slots')">+ New Booking</button></div>

<div class="card"><div class="table-wrap"><table id="bookingTable"><thead><tr><th>Booking ID</th><th>Examination</th><th>Center</th><th>Date</th><th>Time</th><th>Status</th><th>Action</th></tr></thead><tbody id="bookingBody">

<tr><td>BK-10284</td><td>National Aptitude Examination</td><td>Chennai Central</td><td>10 Sep 2026</td><td>09:00–10:00</td><td>CONFIRMED</td><td><button class="btn btn-danger" onclick="cancelBooking(this)">Cancel</button></td></tr>

<tr><td>BK-10251</td><td>CS Fundamentals</td><td>Hyderabad National</td><td>13 Sep 2026</td><td>09:30–11:00</td><td>CONFIRMED</td><td><button class="btn btn-danger" onclick="cancelBooking(this)">Cancel</button></td></tr>

</tbody></table></div></div>

</section>

<section id="attempt" class="page">

<div class="page-head"><div><h1>Exam Attempt</h1><p>National Aptitude Examination · 60 minutes · 10 questions</p></div>ONLINE</div>

<div id="questions"></div>

<div class="card" style="padding:18px;display:flex;justify-content:space-between;align-items:center">All responses are saved for evaluation.<button class="btn btn-success" onclick="submitExam()">Submit Examination</button></div>

</section>

<section id="results" class="page">

<div class="page-head"><div><h1>Result Management</h1><p>View published results and examiner processing status.</p></div></div>

<div class="grid grid-4" style="margin-bottom:20px"><div class="stat"><div><div class="label">Published</div><div class="value">9,742</div></div><div class="stat-icon

green">✓</div></div><div class="stat"><div><div class="label">Pending Review</div><div class="value">183</div></div><div class="stat-icon orange">🟡</div></div><div class="stat"><div><div class="label">Average Score</div><div class="value">72.4%</div></div><div class="stat-icon blue">🟢</div></div><div class="stat"><div><div class="label">Pass Rate</div><div class="value">81.7%</div></div><div class="stat-icon purple">🟣</div></div></div>

<div class="card"><div class="card-head"><h3>Candidate Results</h3><input class="search" placeholder="Search results..." oninput="filterTable(this.value,'resultTable')"></div><div class="table-wrap"><table id="resultTable"><thead><tr><th>Result ID</th><th>Candidate</th><th>Examination</th><th>Score</th><th>Grade</th><th>Status</th><th>Published</th></tr></thead><tbody>

<tr><td>RES-90482</td><td>Harshith Karnam</td><td>National Aptitude Examination</td><td>9 / 10</td><td>A</td><td>PUBLISHED</td><td>02 Sep 2026</td></tr>

<tr><td>RES-90471</td><td>Arun Kumar</td><td>National Aptitude Examination</td><td>8 / 10</td><td>A</td><td>PUBLISHED</td><td>02 Sep 2026</td></tr>

<tr><td>RES-90469</td><td>Priya S</td><td>CS Fundamentals</td><td>24 / 30</td><td>B</td><td>DRAFT</td><td>—</td></tr>

</tbody></table></div></div>

</section>

<section id="analytics" class="page">

<div class="page-head"><div><h1>DBMS Analysis Dashboard</h1><p>File organization, indexing, hashing, query optimization and transaction metrics.</p></div></div>

<div class="grid grid-4" style="margin-bottom:20px"><div class="stat"><div><div class="label">Indexed Queries</div><div class="value">94%</div></div><div class="stat-icon blue">🟢</div></div><div class="stat"><div><div class="label">Avg Lookup</div><div class="value">4.2ms</div></div><div class="stat-icon green">🟢</div></div><div class="stat"><div><div class="label">Transactions</div><div class="value">18.6K</div></div><div class="stat-icon purple">🟣</div></div><div class="stat"><div><div class="label">Rollback Rate</div><div class="value">0.2%</div></div><div class="stat-icon orange">🟡</div></div></div>

<div class="grid grid-2">

<div class="card"><div class="card-head"><h3>Indexing Strategy</h3>B+ TREE</div><div class="card-body"><p style="font-size:13px;margin-bottom:15px">Primary and secondary indexes support candidate-wise attempt history and center/date audit

queries.</p><table><tr><td>idx_attempt_candidate_date</td><td>Candidate +
Date</td><td><span class="badge badge-
green">ACTIVE</td></tr><tr><td>idx_attempt_center_date</td><td>Center +
Date</td><td><span class="badge badge-
green">ACTIVE</td></tr><tr><td>idx_booking_candidate</td><td>Candidate
ID</td><td>ACTIVE</td></tr></table></div></div>

<div class="card"><div class="card-head"><h3>Query Optimization</h3><button class="btn btn-
light" onclick="runExplain()">Run EXPLAIN</button></div><div class="card-body"><div class="code"
id="explainOutput">SELECT a.attempt_id, a.attempt_date, a.score

FROM exam_attempts a

WHERE a.candidate_id = 1

ORDER BY a.attempt_date DESC;

Status: Indexed retrieval

Estimated rows: 4

Access type: ref

Key: idx_attempt_candidate_date</div></div></div>

</div>

</section>

<section id="hashing" class="page">

<div class="page-head"><div><h1>Hash Lookup Demonstration</h1><p>Educational separate-
chaining hash demonstration for candidate lookup.</p></div><span class="badge badge-
purple">HASH TABLE</div>

<div class="card" style="margin-bottom:18px"><div class="card-body"><div
class="code">hash(candidate_id) = (candidate_id × 31) mod 997

bucket = candidate_id mod 5

collision resolution = separate chaining</div></div></div>

<div class="grid grid-3">

<div class="hash-bucket"><h4>BUCKET 0</h4><div class="chain">Candidate #100 · Arun
Kumar</div><div class="chain">Candidate #105 · Priya S</div></div>

<div class="hash-bucket"><h4>BUCKET 1</h4><div class="chain">Candidate #101 · Rahul
M</div></div>

```
<div class="hash-bucket"><h4>BUCKET 2</h4><div class="chain">Candidate #102 · Divya
R</div><div class="chain">Candidate #107 · Karthik P</div></div>
```

```
<div class="hash-bucket"><h4>BUCKET 3</h4><div class="chain">Candidate #103 · Meena
K</div></div>
```

```
<div class="hash-bucket"><h4>BUCKET 4</h4><div class="chain">Candidate #104 · Sanjay
V</div></div>
```

```
</div>
```

```
</section>
```

```
<section id="transactions" class="page">
```

```
<div class="page-head"><div><h1>Transaction & Concurrency Control</h1><p>Demonstration of
ACID-safe slot booking and rollback behavior.</p></div><span class="badge badge-green">ACID
ENABLED</span></div>
```

```
<div class="grid grid-2">
```

```
<div class="card"><div class="card-head"><h3>Slot Booking Transaction</h3></div><div
class="card-body"><div class="code">BEGIN TRANSACTION;
```

```
SELECT * FROM exam_slots
```

```
WHERE slot_id = 101
```

```
FOR UPDATE;
```

```
CHECK available_seats > 0;
```

```
INSERT INTO bookings (...);
```

```
UPDATE exam_slots
```

```
SET available_seats = available_seats - 1;
```

```
COMMIT;</div><div style="margin-top:14px"><button class="btn btn-primary"
onclick="simulateTransaction()">Simulate Transaction</button><button class="btn btn-light"
onclick="notify('SAVEPOINT created — partial changes can be rolled back')">SAVEPOINT
Demo</button></div></div></div>
```

```
<div class="card"><div class="card-head"><h3>Concurrency Protection</h3></div><div class="card-body"><div class="grid grid-2"><div class="kpi">Session A<strong>LOCKED</strong></div><div class="kpi">Session B<strong>WAITING</strong></div></div><hr style="border:0;border-top:1px solid var(--border);margin:20px 0"><p style="font-size:13px;color:var(--muted)">Row-level locking ensures that when two candidates attempt to reserve the final seat, only one transaction can successfully decrement the seat count and create the booking.</p><div class="progress" style="margin-top:15px"><i style="width:99.8%"></i></div><small style="display:block;margin-top:6px;color:#64748b">Consistency guarantee: 99.8% successful transactions in benchmark</small></div></div>
```

```
</div>
```

```
</section>
```

```
<section id="centers" class="page">
```

```
<div class="page-head"><div><h1>Examination Centers</h1><p>Manage authorized examination locations and capacities.</p></div><button class="btn btn-primary" onclick="openModal('centerModal')">+ Add Center</button></div>
```

```
<div class="grid grid-3">
```

```
<div class="card"><div class="card-body"><span class="badge badge-green">ACTIVE</span><h3 style="margin:10px 0 4px;color:var(--dark)">Chennai Central Exam Centre</h3><p style="font-size:12px;color:var(--muted)">Chennai, Tamil Nadu</p><div class="meta" style="margin-top:14px"><span>Capacity <b>100</b></span><span>Occupied <b>24</b></span></div><div class="progress"><i style="width:24%"></i></div></div></div>
```

```
<div class="card"><div class="card-body"><span class="badge badge-green">ACTIVE</span><h3 style="margin:10px 0 4px;color:var(--dark)">Bengaluru Knowledge Centre</h3><p style="font-size:12px;color:var(--muted)">Bengaluru, Karnataka</p><div class="meta" style="margin-top:14px"><span>Capacity <b>80</b></span><span>Occupied <b>26</b></span></div><div class="progress"><i style="width:33%"></i></div></div></div>
```

```
<div class="card"><div class="card-body"><span class="badge badge-green">ACTIVE</span><h3 style="margin:10px 0 4px;color:var(--dark)">Hyderabad National Test Centre</h3><p style="font-size:12px;color:var(--muted)">Hyderabad, Telangana</p><div class="meta" style="margin-top:14px"><span>Capacity <b>90</b></span><span>Occupied <b>27</b></span></div><div class="progress"><i style="width:30%"></i></div></div></div>
```

```
</div>
```

```
</section>
```

```
<section id="users" class="page">
```

```
<div class="page-head"><div><h1>Users & Roles</h1><p>Role-based access control for the examination platform.</p></div><button class="btn btn-primary" onclick="openModal('userModal')">+ Add User</button></div>
```

```
<div class="card"><div class="table-wrap"><table><thead><tr><th>User</th><th>Email</th><th>Role</th><th>Status</th><th>Last Login</th></tr></thead><tbody>
```

```
<tr><td><b>Harshith Karnam</b></td><td>admin@smartexam.local</td><td><span class="badge badge-purple">ADMIN</span></td><td><span class="badge badge-green">ACTIVE</span></td><td>Today, 09:14</td></tr>
```

```
<tr><td><b>Demo Candidate</b></td><td>candidate@example.com</td><td><span class="badge badge-blue">CANDIDATE</span></td><td><span class="badge badge-green">ACTIVE</span></td><td>Today, 08:52</td></tr>
```

```
<tr><td><b>Demo Examiner</b></td><td>examiner@example.com</td><td><span class="badge badge-orange">EXAMINER</span></td><td><span class="badge badge-green">ACTIVE</span></td><td>Yesterday, 18:10</td></tr>
```

```
</tbody></table></div></div>
```

```
</section>
```

```
</main>
```

```
</div>
```

```
</div>
```

```
<div class="modal" id="examModal"><div class="modal-box"><div class="modal-head"><h3>Create Examination</h3><button class="close" onclick="closeModal('examModal')">x</button></div><div class="modal-body"><form onsubmit="createExam(event)" class="form-grid"><div class="field full"><label>Examination Name</label><input required placeholder="e.g. National Aptitude Examination"></div><div class="field"><label>Duration (minutes)</label><input type="number" required value="60"></div><div class="field"><label>Question Count</label><input type="number" required value="30"></div><div class="field full"><label>Description</label><textarea placeholder="Examination description"></div><div class="full"><button class="btn btn-primary">Create Examination</button></div></form></div></div></div>
```

```
<div class="modal" id="centerModal"><div class="modal-box"><div class="modal-head"><h3>Add Examination Center</h3><button class="close" onclick="closeModal('centerModal')">x</button></div><div class="modal-body"><form onsubmit="createCenter(event)" class="form-grid"><div class="field"><label>Center Name</label><input required></div><div class="field"><label>Capacity</label><input type="number" min="1" value="100" required></div><div class="field
```

```
full"><label>Location</label><input required placeholder="City, State"></div><div
class="full"><button class="btn btn-primary">Add Center</button></div></form></div></div></div>
```

```
<div class="modal" id="userModal"><div class="modal-box"><div class="modal-head"><h3>Add
User</h3><button class="close" onclick="closeModal('userModal')">x</button></div><div
class="modal-body"><form onsubmit="createUser(event)" class="form-grid"><div
class="field"><label>Full Name</label><input required></div><div
class="field"><label>Email</label><input type="email" required></div><div
class="field"><label>Role</label><select><option>Candidate</option><option>Examiner</option><o
ption>Admin</option></select></div><div class="field"><label>Password</label><input
type="password" required></div><div class="full"><button class="btn btn-primary">Create
User</button></div></form></div></div></div>
```

```
<div class="toast" id="toast"></div>
```

```
<script>
```

```
const questions=[
```

```
["What is the main purpose of the attention mechanism?","To focus on the most relevant parts of
the input","To remove all input information","To reduce the number of classes","To increase image
size"],0],
```

```
["In self-attention, the model mainly relates:","Only the first and last elements","Only the input and
output images","Different training datasets","Different parts of the same input sequence"],3],
```

```
["What is the purpose of masked self-attention?","To prevent a position from attending to future
positions","To remove all attention scores","To increase the image resolution","To remove the input
sequence"],0],
```

```
["Multi-head self-attention uses:","Multiple input images only","Only one attention score","Multiple
attention heads to capture different relationships","No attention mechanism"],2],
```

```
["A Transformer is mainly based on:","Pooling only","Attention mechanisms","Decision trees","K-
means clustering"],1],
```

```
["How does a Vision Transformer (ViT) process an image?","It uses only pooling layers","It converts
the image directly into audio","It removes all image information","It divides the image into patches
and processes them as tokens"],3],
```

```
["One major difference between a Transformer and a CNN is that:","Transformers use attention to
model relationships between input elements","Transformers cannot process images","CNNs do not
have trainable parameters","CNNs can only process text"],0],
```

```
["What is the main role of the Transformer encoder?","To increase image resolution","To delete the
input sequence","To transform the input into useful contextual representations","To store training
data"],2],
```

```
["What is the purpose of input embedding?","To convert input tokens into numerical vector representations","To remove the input tokens","To increase the number of images","To calculate the final accuracy"],0],
```

```
["Why is positional encoding used in a Transformer?","To remove the input sequence","To provide information about the position or order of input elements","To reduce the number of attention heads","To increase the image resolution"],1]
```

```
];
```

```
function showPage(id,btn){
```

```
    document.querySelectorAll('.page').forEach(p=>p.classList.remove('active'));
```

```
    const el=document.getElementById(id); if(el)el.classList.add('active');
```

```
    document.querySelectorAll('.nav button').forEach(b=>b.classList.remove('active'));
```

```
    if(btn)btn.classList.add('active');
```

```
    const title=id.charAt(0).toUpperCase()+id.slice(1);
```

```
    document.querySelector('.breadcrumb').innerHTML='<b>Smart National Exam System</b> / '+title;
```

```
    document.getElementById('sidebar').classList.remove('open');
```

```
    window.scrollTo({top:0,behavior:'smooth'});
```

```
}
```

```
function toggleSide(){document.getElementById('sidebar').classList.toggle('open')}
```

```
function notify(msg){const
```

```
t=document.getElementById('toast');t.textContent=msg;t.classList.add('show');setTimeout(()=>t.classList.remove('show'),2600)}
```

```
function openModal(id){document.getElementById(id).classList.add('show')}
```

```
function closeModal(id){document.getElementById(id).classList.remove('show')}
```

```
document.querySelectorAll('.modal').forEach(m=>m.addEventListener('click',e=>{if(e.target===m)m.classList.remove('show')}));
```

```
function filterSlots(q){q=q.toLowerCase();document.querySelectorAll('#slotGrid .slot').forEach(x=>x.style.display=x.dataset.search.includes(q)?'block':'none')}
```

```
function filterTable(q,id){q=q.toLowerCase();document.querySelectorAll('#'+id+' tbody tr').forEach(x=>x.style.display=x.innerText.toLowerCase().includes(q)?'table-row':'none')}
```

```
function bookSlot(exam,center,date,time){
```



```

const id='BK-'+Math.floor(10000+Math.random()*89999);

const tr=document.createElement('tr');

tr.innerHTML=`<td>${id}</td><td>${exam}</td><td>${center}</td><td>${date}</td><td>${time}</td>
<td><span class="badge badge-green">CONFIRMED</span></td><td><button class="btn btn-
danger" onclick="cancelBooking(this)">Cancel</button></td>`;

document.getElementById('bookingBody').prepend(tr);notify('Slot booked successfully —
'+id);showPage('bookings');

}

function cancelBooking(btn){btn.closest('tr').querySelector('td:nth-child(6)').innerHTML='<span
class="badge badge-red">CANCELLED</span>';btn.remove();notify('Booking cancelled and seat
released')}

function startExam(){showPage('attempt');renderQuestions();notify('Exam attempt loaded')}

function renderQuestions(){

const box=document.getElementById('questions');if(box.dataset.loaded)return;

box.dataset.loaded='1';

box.innerHTML=questions.map((q,i)=>`<div class="question"><h3>Q${i+1}.
${q[0]}</h3>${q[1].map((o,j)=>`<label class="option"><input type="radio" name="q${i}"
value="${j}">${o}</label>`).join("")}</div>`).join("");

}

function submitExam(){

let score=0,answered=0;

questions.forEach((q,i)=>{const
a=document.querySelector(`input[name="q${i}"]:checked`);if(a){answered++;if(+a.value===q[2])score
++}});

if(answered<questions.length){notify('Please answer all 10 questions before submitting');return}

notify('Exam submitted successfully. Score: '+score+'/10');setTimeout(()=>showPage('results'),700);

}

function runExplain(){document.getElementById('explainOutput').textContent+='\n\nEXPLAIN
ANALYZE completed.\nActual rows: 4\nExecution time: 0.84 ms\nIndex used:
idx_attempt_candidate_date\nOptimization: SUCCESS';notify('EXPLAIN analysis completed')}

function simulateTransaction(){notify('BEGIN → row locked → booking inserted → seat decremented
→ COMMIT')}

```

```
function createExam(e){e.preventDefault();closeModal('examModal');notify('Examination created successfully')}
```

```
function createCenter(e){e.preventDefault();closeModal('centerModal');notify('Examination center added successfully')}
```

```
function createUser(e){e.preventDefault();closeModal('userModal');notify('User created with role-based access')}
```

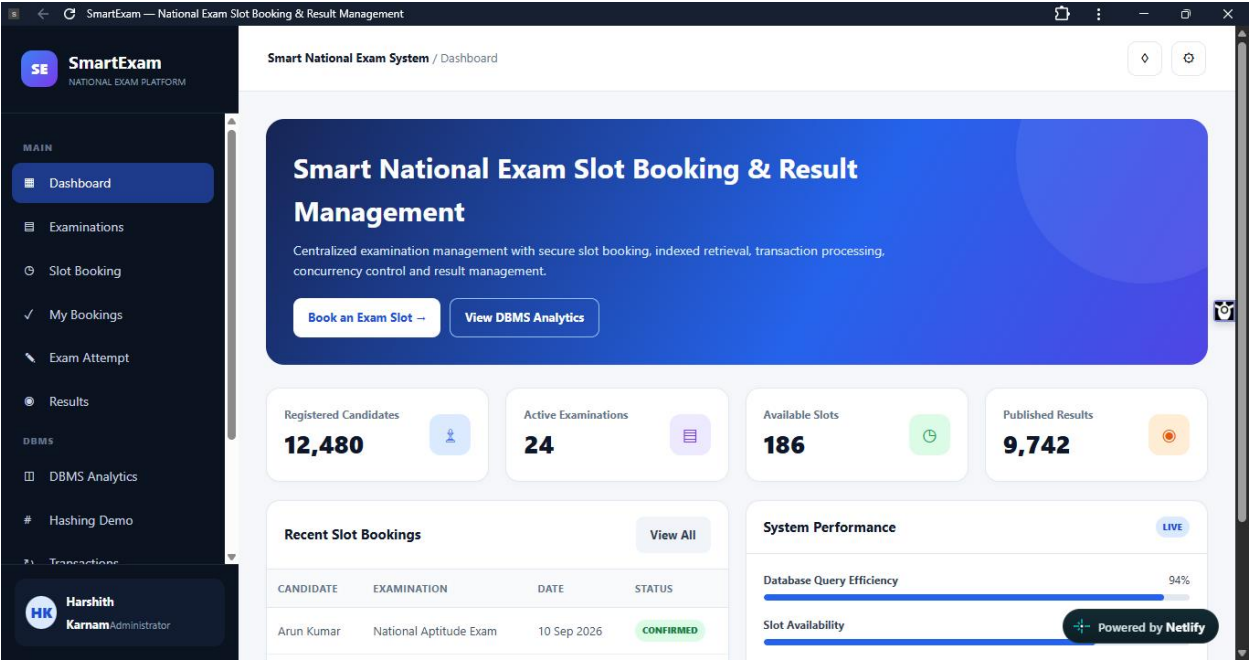
```
renderQuestions();
```

```
</script>
```

```
</body>
```

```
</html>
```

OUTPUT DASHBOARD



INPUT

